JOÃO VITOR FURTADO DE SOUSA

Software Engineering Intern · Computational Engineering @ UFJF

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SUMMARY

Computational Engineering student (UFJF, graduating Dec 2028) working as a software engineering intern on aviation-infrastructure systems. Rewrote a production camera-monitoring service from Python to C++, cutting server CPU load from 80–90% on an Intel i9 to roughly 7% on an i3. Comfortable across backend development, systems programming and applied AI; ICPC Brazilian Regional competitor. Fluent in English, open to remote and international roles.

EXPERIENCE

Software Engineering Intern — AMD Services Juiz de Fora, Brazil · 2025–Present

- Rebuilt a PTZ camera computer-vision monitoring system from Python to C++, cutting server CPU load from **80–90% on an Intel i9 to roughly 7% on an i3** — letting the workload run on far cheaper hardware.
- Cycles the camera through nine presets in a continuous loop, streaming a live preview and capturing the final frame of each position into a 9-image mosaic, alongside fisheye sky-camera and weather-station feeds.
- Designed and built a staff management platform from scratch (Spring Boot, REST, SQL), covering vacation scheduling, duty rosters, aeromedical certification tracking and automated email notifications for personnel across multiple air-traffic service stations.
- Maintained and extended production aviation systems: an airport billing platform, a Radio-over-IP (RoIP) system, an occurrence-logging system and the institutional website.

IT Assistant (Scholarship) — IF Sudeste MG Juiz de Fora, Brazil · 2022

- Diagnosed and repaired computer peripherals and electronic components.

Undergraduate Research Scholar — PIC / OBMEP 2018–2022

- Five consecutive years in OBMEP's scientific initiation programme (PIC), studying applied mathematics.

PROJECTS

Q-Learning Investment Policy — *Python, Gymnasium*

Custom RL environment and state-discretisation pipeline over PETR4 market data to learn a trading policy.

Guará — Self-Hosted AI Assistant — *Python, Gemini function calling, MCP, openWakeWord*

Open-source (MIT) personal assistant running 24/7 on a home Linux machine, reached via Telegram or a custom "Ô, Guará" wake word trained and calibrated from scratch (93.8% recall on clean audio, ~2 false alarms/hour). Bounded function-calling loop over 17 tool modules, MCP bridge, Gemini model fallback, and confirmation gates before destructive actions.

Smart Desk AI — *Python, OpenCV, ArUco, AI agents*

Workspace-monitoring platform running entirely on a repurposed smartphone: detects desk boundaries via ArUco markers, calibrates real-world scale, identifies objects and exposes that spatial context to AI agents and smart-home devices.

EDUCATION

B.Sc. Computational Engineering — UFJF · 2024 – Dec 2028 (expected) · currently 6th semester

Technical Diploma in Electrotechnics — IF Sudeste MG · 2021–2023

COMPETITIVE PROGRAMMING & AWARDS

- ICPC Brazilian Regional (Maratona de Programação da SBC) — competitor, 2025 and 2026.
- Maratona Mineira de Programação — competitor in every edition, 2025 and 2026.
- Hackathon RERUM 2026 (28th UFJF Computing Week) — winner with CochoCerto, an expert system for dairy herd diets, as a team.
- OBMEP — Brazil's national mathematics olympiad (18M+ students in the first round): **Regional Gold** (2023), Bronze (2019), Silver (2018), Honourable Mention (2017).
- ONC — Brazilian National Science Olympiad: **Gold Medal** (2021).

SKILLS

Languages: C++, Python, Java

Backend & Data: Spring Boot, REST APIs, SQL

AI & Computer Vision: OpenCV, object detection, LLM function calling, MCP, AI agents

Infrastructure: Docker, Linux, AWS, Firebase, Git

Spoken languages: Portuguese (native), English (advanced)